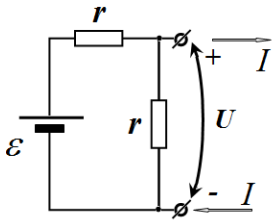
문제 1B (5점)

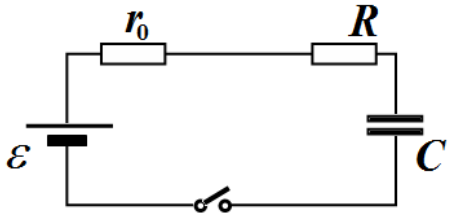
첫 번째 풀이.

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(U I) .



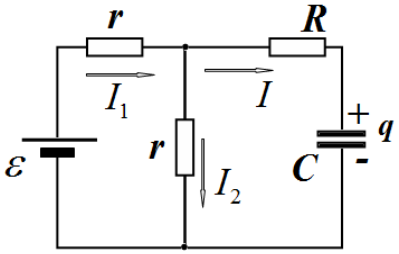
$$\varepsilon_0 = \frac{\varepsilon}{2},\ r_0 = \frac{r}{2}.$$



$$Q_0 = \frac{C\varepsilon^2}{2}. \qquad R$$

$$Q = \frac{R}{R+r_0}Q_0 = \frac{R}{R+\frac{r}{2}}\cdot\frac{C\left(\frac{\varepsilon}{2}\right)^2}{2} = \frac{RC\varepsilon^2}{4(2R+r)}$$

두 번째 풀이.



$$\begin{cases} \varepsilon = I_1 r + I_2 r \\ I_2 r = IR + \dfrac{q}{C} \\ I_1 = I + I_2 \\ I = \dot{q} \end{cases}$$

I_1 I_2

$$I\left(R+\frac{r}{2}\right)+\frac{q}{C}=\frac{\varepsilon}{2},$$

I

$$I^2\left(R+\frac{r}{2}\right)=\frac{\varepsilon}{2}I-\frac{qI}{C}=\frac{\varepsilon}{2}\dot{q}-\frac{2q\dot{q}}{2C}=\frac{d}{dt}\left(\frac{\varepsilon q}{2}-\frac{q^2}{2C}\right).$$

$$I^2R=\frac{R}{R+\frac{r}{2}}\frac{d}{dt}\left(\frac{\varepsilon q}{2}-\frac{q^2}{2C}\right),$$

$$\int_0^\infty I^2R\,dt=\frac{R}{R+\frac{r}{2}}\left(\frac{\varepsilon q}{2}-\frac{q^2}{2C}\right)\Bigg|_{q(0)}^{q(\infty)}$$

$$q(0)=0 \qquad q(\infty)=C\frac{\varepsilon}{2}$$